

Criteria: Optional Criteria 4 — Academic Contributions through Research

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OC4-1: Blind-Spot Decomposition Theory (BSDT) — Research Contribution

1. Executive Summary

Blind-Spot Decomposition Theory (BSDT) is an independently developed analytical framework for diagnosing why an AI fraud-detection system misses the specific transactions it fails to catch. Conventional measures — accuracy, precision, recall — tell you how often a system fails, not why. BSDT decomposes each failure into one of four distinct, measurable causes (camouflage, information gap, unseen pattern, outdated baseline) and produces a single correction score (MFLS) that can be applied on top of any existing detector without rebuilding it. Developed without university affiliation, co-authors, or external funding, BSDT has since been independently adopted into postgraduate course reading at Abiola Ajimobi Technical University (Nigeria) by Dr Sunday Adeola Ajagbe — evidenced by his signed 3-page reference letter accompanying this submission, with publication record in OC4-2.

2. BSDT Framework — Four-Channel Decomposition

The framework analyses each transaction through four independent channels. Each channel produces a score between 0 and 1 (higher = greater blind-spot risk for that cause). The four scores are then averaged to produce the final correction value:

- Channel 1 — Camouflage (δ_C): measures how closely the transaction resembles known fraud patterns. A score near 1 indicates the transaction looks suspicious to an informed examiner; a score near 0 means it blends in with normal activity.
- Channel 2 — Information Gap (δ_G): measures how much of the relevant evidence the system is missing. A high score indicates the system is making a decision with incomplete information.
- Channel 3 — Unseen Pattern (δ_A): measures how different this transaction's behaviour is from anything the system learned from during training. A high score means the system has no reliable reference for this type of activity.
- Channel 4 — Outdated Baseline (δ_T): measures how much the general pattern of transactions has changed since the system was last calibrated. A high score indicates the system's reference point may no longer reflect current conditions.

Formal definitions (each channel produces a score bounded between 0 and 1; the four scores are combined equally to form the final MFLS correction value):

$\delta_C = 1 - \text{clip}(\ x - \mu_{\text{ref}}\ / d_{\text{max}}, 0, 1)$	(Camouflage)
$\delta_G = (\# \text{ near-zero features}) / p$	(Feature Gap)
$\delta_A = \sigma((d_{\text{Mahal}} - d_{\text{ref}}) / d_{\text{ref}})$	(Activity Anomaly)
$\delta_T = \sigma(0.5 \cdot (d_{\text{kNN}} / d_{\text{kNN,ref}} - 2))$	(Temporal Novelty)
$E_{\text{BS}} = \sum \delta_i^2$	(Total blind-spot energy)
$\text{MFLS} = \ \nabla E_{\text{BS}}\ _F$	(Gradient-norm score)
$\text{Combined} = 0.5 \cdot \hat{E}_{\text{BS}} + 0.5 \cdot \text{MFLS}$	(Equal-weight average)

Table 1: The mathematical definition of each channel. Each score (δ) is expressed on a 0-to-1 scale, where 1 indicates maximum blind-spot risk for that channel. The final MFLS correction is the equal-weight average of all four channel scores.

3. Code Implementation

BSDT is implemented as a software component called BSDTChannels, available in the public code repository (system_mode.py). It exposes three straightforward operations:

- The component exposes three operations: fit(X_ref) calibrates the framework on a reference set of normal transactions (one-time setup); mfls(X) computes the Missed-Fraud Likelihood Score for a batch of new transactions; score(X) returns the final 0-to-1 correction value, ready to be applied on top of any existing fraud detector.

```
class BSDTChannels:
    """Blind-Spot Detection Tensor — four complementary channels."""

    def fit(self, X_ref: np.ndarray) -> "BSDTChannels":
        self.mu_ = X_ref.mean(axis=0)
        self.d_max_ = max(float(np.linalg.norm(X_ref - self.mu_, axis=1).max()), self.eps)
        self.mahal_ref_median_ = float(np.median(self._mahalanobis(X_ref)))
        self.ref_knn_median_ = float(np.median(ref_dists[:, -1]))
```

```
def mfls(self, X: np.ndarray) -> np.ndarray:
    return np.linalg.norm(self._gradient_vectors(X), axis=1)

def score(self, X: np.ndarray) -> np.ndarray:
    e = self.energy(X)
    m = self.mfls(X)
    return 0.5 * (e / e.max()) + 0.5 * (m / m.max())
```

Figure 1: The production code for BSDTChannels, taken directly from the public repository. The three operations — setup, scoring, and output — correspond to the steps described above.

4. Validation Results

BSDT was first tested on its primary intended domain: detecting financial fraud. In a second stage, the same framework was applied — as an exploratory extension — to structurally similar problems in other fields, to test whether the four-channel approach had broader mathematical relevance beyond fraud detection. The results are summarised in the table below.

Dataset / Domain	Evidence	Key metric
Fraud benchmark (primary)	99.94% false-positive reduction	36,469 → 22 alerts
12 fraud datasets (66 variants)	Zero missed fraud in 10 of 12 datasets	Mean F1 0.787
2021 Texas ERCOT power grid (later extension)	Detected instability signals associated with the 2021 grid stress event 72 hours in advance	AUC 0.9997 (public data)
2022 Terra/Luna (later extension)	Detected on-chain anomaly signals 30 hours preceding the de-pegging event	On-chain data only
2008 US banking crisis (later extension)	Early-warning signals consistent with known institution failures	AUROC 0.867

Table 2: Validation results across five test scenarios. The first two rows are the primary validation (financial fraud detection). The final three rows show the same framework applied to structurally similar problems in other fields. All data used is publicly available.

The fraud-detection results (first two rows) are the core validation evidence. The three cross-domain rows show the same framework independently identifying early-warning signals in power-grid stress and financial crisis data. This is not because those domains were specifically targeted, but because they share the same underlying mathematical structure as fraud detection: things that look normal on the surface, unusual activity signals underneath, and reference points that become outdated over time. These cross-domain extensions are formalised as standalone contributions in the ‘Geometry of System Collapse’ and ‘Universal Deviation Principle’ papers (see OC4-2, Sections 4 and 5). Only publicly available data was used throughout; no private or privileged information was accessed at any stage.

Independent validation — no data leakage: all 12 fraud benchmarks were evaluated using strict time-based splits, meaning the model was trained only on data from before the test period. A dedicated verification script (test_no_leakage_all.py) confirms that no data contamination occurred across any of the 12 datasets.

5. Reproducibility

BSDT was evaluated across 792 separate test runs: 66 model configurations, each tested on 12 independent fraud datasets. This scale rules out the possibility that a favourable result was a coincidence or a cherry-picked example. Every test script is publicly available and can be run on a standard laptop in 2–8 minutes using free Python libraries. No specialist hardware is required.

Script	What it tests	Result
test_no_leakage_all.py	Zero data leakage across all 12 fraud datasets	Pass — all temporal splits clean
test_mode4.py	BSDT blind-spot correction on fraud benchmarks	99.94% FP reduction, 0 missed fraud
test_mode6_bench.py	66-variant systematic comparison	Best model wins 10/12 datasets
test_molecular_bench.py	Cross-domain: molecule property prediction	BSDT improves MAE on 4/6 targets
test_geo_cyber.py	Cross-domain: cyber threat + geopolitical risk	AUC improvement vs baseline
run_bsd_t_benchmarks.py	Full 792-evaluation orchestration script	All 66 variants, 12 datasets

Table 3: The six test scripts covering the full 792-evaluation study. Each can be run independently by any assessor with a standard Python installation. The scale of testing means no single result can be dismissed as coincidental or cherry-picked.

```
# Run full BSDT benchmark suite (792 evaluations)
# From repository root:
py -3 run_bsd_t_benchmarks.py
```

```
# Reproduce the key fraud result:
py -3 test_mode4.py

# Verify zero data leakage:
py -3 test_no_leakage_all.py

# Cross-domain: ERCOT power grid + Terra/Luna crypto:
py -3 test_geo_cyber.py
```

Figure 2: The commands required to reproduce the results from a standard Python 3.11+ installation, run from the public code repository.

6. Technical Significance

- BSDT offers two practical contributions to deployed fraud-detection systems. First, it explains why a detection failed by attributing it to one of four specific mechanisms — giving operational teams actionable diagnostic information rather than an unexplained miss. Second, the correction score is additive: it sits alongside any existing fraud model without retraining or replacing it, so deployed infrastructure benefits immediately.

Theoretical extensions: Geometry of System Collapse and the Universal Deviation Law (UDL)

BSDT's four-channel decomposition raises a natural theoretical question: are four channels the right number, and what is the mathematical structure that makes the decomposition work? This question is addressed in two follow-up papers.

Geometry of System Collapse (Zenodo, March 2026) gives the four BSDT channels a geometric reading: each channel becomes one axis in a small coordinate space, and a system drifting toward failure traces a visible path through that space. The picture is the same one a structural engineer uses when watching stress build up in a bridge before it gives way. The paper applies this view to public power-grid stress data and cryptocurrency crisis data, and shows that the same early-warning signal that flags fraudulent transactions also flags these structurally similar failures.

The Universal Deviation Law (UDL, also called the Universal Deviation Principle on Zenodo) generalises the four-channel idea. Instead of fixing the number of channels at four, UDL lets the analyst combine any number of independent "detectors," each one measuring abnormal behaviour in its own way. The paper shows three things in plain terms: (1) every detector can be scored for how much real information it adds; (2) those scores tell us the best way to weight the detectors when combining them; and (3) once they are combined this way, any genuine anomaly above a clearly stated size is guaranteed to be caught. BSDT is then the special case where exactly four detectors are used, and UDL is the wider rule the same logic extends to.

Both extension papers are published on Zenodo with permanent DOIs; full bibliographic details and view/download metrics are in OC4-2.

BSDT was developed entirely independently, published openly without any institutional support, and has been incorporated into postgraduate course reading at Abiola Ajimobi Technical University. The full publication record — including the two theoretical extensions above — is documented in OC4-2; the academic adoption is evidenced by the accompanying 3-page reference letter from Dr Sunday Adeola Ajagbe.